

Manual

EN

Handleiding

NL

Manuel

FR

Anleitung

DE

Manual

ES

Användarhandbok

SE

Appendix

Phoenix Inverter Smart

12 | 1600 230V

12 | 2000 230V

12 | 3000 230V

24 | 1600 230V

24 | 2000 230V

24 | 3000 230V

48 | 1600 230V

48 | 2000 230V

48 | 3000 230V

1. SAFETY INSTRUCTIONS

General

Please familiarize yourself with the safety features and instructions by first reading the documentation supplied with this product before using the equipment. This product has been designed and tested in accordance with international standards. The equipment must be used exclusively for the purpose for which it was designed.

WARNING: ELECTRIC SHOCK HAZARD.

The product is used in combination with a permanent energy source (battery). Even if the equipment is switched off, a dangerous electrical voltage can occur at the input and/or output terminals. Always disconnect the battery before performing maintenance.

The product has no internal user-serviceable components. Do not remove the front plate or operate the product if any panels have been removed. All servicing must be undertaken by qualified personnel.

Never use the product where there is a risk of gas or dust explosions. Consult the battery manufacturer's information to ascertain that the product is intended for use in conjunction with the battery. Always comply with the battery manufacturer's safety instructions.

WARNING: Do not lift heavy loads without assistance.

Installation

Read the installation instructions in the installation manual before installing the equipment.

This is a Safety Class I product (supplied with a protective grounding terminal). **The chassis must be grounded.** A grounding point is located on the outside of the product. Whenever it is likely that the grounding protection has been damaged, the product must be turned off and secured against unintended operation; please contact qualified service staff.

Ensure that the DC and AC input cables are fused and fitted with circuit breakers. **There is no internal fuse inside this product.** Never replace a safety component with a different type. Consult the manual to determine the correct component.

During installation ensure that the remote connector with wire bridge is removed (or switch off the remote on/off switch if installed) to be sure that the inverter cannot be switched on unexpectedly.

Before applying power, ensure that the available power source matches the configuration settings of the product as described in the manual.

Ensure that the equipment is used under the correct ambient conditions. Never operate the product in a wet or dusty environment. Ensure there is adequate free space for ventilation around the product and check that the ventilation vents are not blocked.

Ensure that the required system voltage does not exceed the product's capacity.

Transport and Storage

Ensure that the mains power and battery leads have been disconnected before storing or transporting the product.

No liability can be accepted for any transport damage if the equipment is shipped in non-original packaging.

Store the product in a dry environment; the storage temperature must be between -20°C and 60°C.

Consult the battery manufacturer's manual in respect of transport, storage, charging, recharging and disposal of the battery.

2. DESCRIPTION

2.1 General

Bluetooth built-in: fully configurable with a tablet or smartphone

- Low battery voltage alarm trip and reset levels
- Low battery voltage cut-off and restart levels
- Dynamic cut-off: load dependent cut-off level
- Output voltage: 210 - 245V
- Frequency: 50 Hz or 60 Hz
- ECO mode on/off and ECO mode sense level
- Alarm relay

Monitoring:

- In- and output voltage, % load and alarms

VE.Direct communication port

The VE.Direct port can be connected to a computer (VE.Direct to USB interface cable needed) to configure and monitor the same parameters.

Proven reliability

The full bridge plus toroidal transformer topology has proven its reliability over many years.

The inverters are short circuit proof and protected against overheating, whether due to overload or high ambient temperature.

High start-up power

Needed to start loads such as power converters for LED lamps, halogen lamps or electric tools.

ECO mode

When in ECO mode, the inverter will switch to standby when the load decreases below a preset value (min load turn on level: 10VA; and min load turn off level: 0VA). Once in standby the inverter will switch on for a short period (adjustable, default: every 3 seconds). If the load exceeds a preset level, the inverter will remain on.

Remote on/off

A remote on/off switch can be connected to a two pole connector, between battery plus and the left hand contact of the two pole connector or between battery minus and the right hand contact of the two pole connector

LED diagnosis

See section 3.3

To transfer the load to another AC source: the automatic transfer switch

For our low power inverters we recommend our Filax Automatic Transfer Switch. The Filax features a very short switchover time (less than 20 milliseconds) so that computers and other electronic equipment will continue to operate without disruption. Alternatively use a MultiPlus with built-in transfer switch.

3. OPERATION

3.1 On/Off Switch

When switched to "on" with the push button, the product is fully functional. The inverter will come into operation and the LED "inverter" will light up. By pushing the push button subsequently, within a short period of time, the inverter toggles between "on", "ECO" and "off".

Apart from the pushbutton; the inverter can also be switched on (normal or ECO) and off with Bluetooth on a mobile device running iOS or Android and the Victron Connect app. However when switched off via Bluetooth or the push button; the unit **cannot** be switched on and off again via the wired VE.Direct port.

3.2 Remote control

Remote control is possible with a simple on/off switch or with a Phoenix Inverter Control panel. A switch for remote control (on/off) can be connected to a two pole connector. The switch can also be connected between battery plus and the left hand contact of the two pole connector (marked "H"; see appendix A) or between battery minus and the right hand contact of the two pole connector (marked "L"; see appendix A).



For safety purposes, this product can be turned off completely (i.e. the inverter cannot be switched on via the push button or Bluetooth) by removing the remote connector and its default installed wire bridge (or switch off the remote on/off switch if installed). The user can then be certain that the inverter cannot be switched on accidentally via Bluetooth by an unexpected other user.

3.3 LED definitions

Green LED	Status	Trouble shooting
 Solid on	Inverter on	Red LED Off status OK Red LED On or blinking: The Inverter is still on, but will shut down when the condition gets worse. See red LED table for warning reason
 Slow single pulse	ECO mode	If the inverter keeps switching on and off while there is a load connected, the load may be too small compared to the actual ECO mode settings. Increase the load or change ECO mode settings. (minimum ECO mode setting: 15W)
 Fast double pulse	Off and waiting	Inverter did shut down because of a protection. The inverter will restart automatically as soon as all alarm conditions are cleared. See red LED state for the shutdown reason.
 Off	Inverter off	Red LED Off Check Remote on/off connector. Check DC cable connections and fuses. Check operational mode by pushing push button one time. Red LED On or blinking The inverter did shut down because of a protection. It will no longer automatically restart. The red LED indicates the reason for shutdown. Remove the cause and then restart the inverter by switching it Off, and then back On.
 Fast	Off and	Red LED Blinking (-●-●-●-●-)



blink	firmware update in progress or failed	Firmware update in progress or firmware update failed. When failed; retry firmware update.
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Yellow LED	Status	Trouble shooting
●●●●●●●● Solid on	ECO mode	Red LED Off status OK Red LED On or blinking: The inverter is still on, but will shut down when the condition gets worse. See red LED table for warning reason
----- Off	ECO mode off	Red LED Off Check operational mode by pushing push button one time. Check Remote on/off connector. Check DC cable connections and fuses. Red LED On or blinking The inverter did shut down because of a protection. It will no longer automatically restart. The red LED indicates the reason for shutdown. Remove the cause and then restart the inverter by switching it Off, and then back On.

Red LED	Definition	Trouble shooting
●●●●●●●● Solid on	Overload	Reduce load
●●●●----- Slow blink	Low batt.	Recharge or replace battery Check DC cable connections Check cable cross section as it may be insufficient. See section 4.2 Protections and automatic restarts for manual and automatic restart behavior.
●-●-●-●- Fast blink	High batt.	Reduce DC input voltage, check for faulty charger
●-●------ Double pulse	High temp.	Reduce load and/or move inverter to better ventilated area
●----- Fast single pulse	High DC ripple	Check DC cable connections and cable cross section.

3.4 Protections and automatic restarts

Overload

Some loads like motors or pumps draw large inrush currents in a start-up situation. In such circumstances, it is possible that the start-up current exceeds the over current trip level of the inverter. In this case the output voltage will quickly decrease to limit the output current of the inverter. If the over current trip level is continuously exceeded, the inverter will shut down: wait 30 seconds and then restart.

After three restarts followed by another overload within 30 seconds of restarting, the inverter will shutdown and remain off. The LEDs will signal shutdown due to overload. To restart the inverter, switch it Off, then On.

Low battery voltage (adjustable)

The inverter will shut down when the DC input voltage drops below the low battery shutdown level. After a minimum delay of 30 seconds, the inverter will restart if the voltages rise above the low battery restart level.

After three restarts followed by a low battery shutdown within 30 seconds of restarting, the inverter will shutdown and stop retrying. The LEDs will signal low battery shutdown. To restart the inverter, switch it Off, and then On, or recharge the battery: as soon as the battery has risen and then stays above the Charge detect level for 30 seconds, it will switch on.

See the Technical Data table for default low battery shutdown and restart levels. They can be changed with the VictronConnect App.

High battery voltage

Reduce DC input voltage and/or check for a faulty battery- or solar-charger in the system. After shutting down due to a high battery voltage, the inverter will first wait 30 seconds and then retry operation as soon as the battery voltage has dropped to acceptable level. The inverter will not stay off after multiple retries.

High temperature

A high ambient temperature or enduring high load may result in shut down to over temperature. The inverter will restart after 30 seconds. The inverter will not stay off after multiple retries. Reduce load and/or move inverter to better ventilated area.

High DC ripple

High DC ripple is usually caused by loose DC cable connections and/or too thin DC wiring. After the inverter has switched off due to high DC ripple voltage, it waits 30 seconds and then restarts.

After three restarts followed by a shutdown due to high DC ripple within 30 seconds of restarting, the inverter will shutdown and stops retrying. To restart the inverter, switch it Off and then On.

Continuous high DC ripple reduces the inverter life expectancy.

4. INSTALLATION



This product should be installed by a qualified electrician.



During installation ensure that the remote connector with wire bridge is removed (or switch off the remote on/off switch if installed) to be sure that the inverter cannot be switched on unexpectedly.

4.1 Location

The product must be installed in a dry and well-ventilated area, as close as possible to the batteries. There should be a clear space of at least 10cm around the appliance for cooling.



Excessively high ambient temperature will result in the following:
 Reduced service life.
 Reduced charging current.
 Reduced peak capacity, or shutdown of the inverter.
 Never mount the appliance directly above the batteries.

The product is suitable for wall mounting. For mounting see appendix A. The appliance can be mounted horizontally as well as vertically; vertical mounting is preferable. The vertical position offers optimum cooling.



The interior of the product must remain accessible after installation.

Try and keep the distance between the product and the battery to a minimum in order to minimize cable voltage losses.



For safety purposes, this product should be installed in a heat-resistant environment if it is used with equipment where a substantial amount of power is to be converted. You should prevent the presence of e.g. chemicals, synthetic components, curtains or other textiles, etc., in the immediate vicinity.

4.2 Connection of Battery cables

In order to fully utilize the full capacity of the product, batteries with sufficient capacity and battery cables with sufficient cross section should be used. See table.

	12/1600	24/1600	48/1600	12/2000	24/2000	48/2000
Recommended cross section (mm ²)						
length up to 6 m	50	25	25	70	35	25

	12/3000	24/3000	48/3000
Recommended cross section (mm ²)			
0 – 5 m	95	50	35
5 -10 m	120	95	70

	12/1600	24/1600	48/1600	12/2000	24/2000	48/2000
Recommended battery capacity (Ah)	300 - 800	150 - 400	75 - 200	350 - 1000	200 - 500	100 - 250

	12/3000	24/3000	48/3000
Recommended battery capacity (Ah)	400 - 1200	200 - 700	100 - 400

Remark: Internal resistance is the important factor when working with low capacity batteries. Please consult your supplier or the relevant sections of our book "electricity on board", downloadable from our website.

Procedure

Proceed as follows to connect the battery cables:



Use an insulated box spanner in order to avoid shorting the battery.
Avoid shorting the battery cables.

Connect the battery cables: the + (red) and the - (black), to the battery see appendix A. Reverse polarity connection (+ to – and – to +) will cause damage to the product. Secure the nuts tightly in order to reduce the contact resistance as much as possible.